

Zachary Burton

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EDUCATION

Massachusetts Institute of Technology (MIT)

B.S. in Mathematics (GPA: 4.6/5.0)

Cambridge, MA

Sep 2022 – Feb 2026

Coursework: Probability, Statistics, Real Analysis, Linear Algebra, Abstract Algebra, Reinforcement Learning (Grad), Program Synthesis (Grad)

PUBLICATIONS

Structured Hints for Sample-Efficient Lean Theorem Proving | [arXiv:2601.16172](https://arxiv.org/abs/2601.16172)

Poissonization-based collision threshold derivation for random walks on lattices | [arXiv:2505.02973](https://arxiv.org/abs/2505.02973)

EXPERIENCE

Meta

Machine Learning Engineer

Feb 2026 – Present

New York, NY

- Designed offline experiments to measure revenue impact of ranking signal changes in the Reels late-stage pipeline; instrumented metrics to surface tradeoffs between revenue lift and engagement quality
- Proposed architectural change to the core value model to enable more granular signal-level debugging and backtesting; designed evaluation spec and presented to team

Software Engineer Intern

May 2025 – Aug 2025

- Refactored copyright enforcement engine processing 2B+ daily uploads; achieved 97% consistency with legacy outputs, verified via SQL logging and automated regression suite

MIT Computer Science & Artificial Intelligence Lab

Research Engineer

Nov 2024 – Aug 2025

Cambridge, MA

- Built **PIEFACE**, an end-to-end human feedback platform: annotation web UI, behavioral trace parser, and SQL data pipeline for collecting, storing, and joining labeled human preference data at scale
- Designed grading and evaluation infrastructure to measure alignment between learned reward signals and human judgments; integrated reward model training into distributed PPO rollouts in a custom simulation environment
- Formulated gadget simulation search (automating NL-completeness reductions) as an MDP over finite-automaton structures; Maskable PPO achieves 85% success in <3ms vs. 2% random and >100s exhaustive baselines (advised by Jayson Lynch & Erik Demaine)

DISTINGUISHING HONORS

British Physics Olympiad (**Gold, 2022**)

University of Cambridge – Isaac Physics Diamond Award (**top 0.004% nationally, 2021**)

RESEARCH

Structured Hints for Neural Theorem Proving

Independent project | [arXiv:2601.16172](https://arxiv.org/abs/2601.16172)

2025 – 2026

- Diagnosed mode collapse in RL-trained theorem provers: i.i.d. sampling with DeepSeek-Prover-V1.5-RL on miniF2F plateaus at $k=32$ (zero new proofs from $k=32 \rightarrow 64$), indicating severe underexploration of the tactic space
- Designed a fixed schedule of 15 tactic skeletons as inference-time structural guidance; achieved 22.5% pass@16 vs. 15.6% baseline (45% relative improvement) at the same sample budget with no retraining
- Ran controlled diversity ablation separating structural guidance from prompt perturbation: paraphrases match baseline while irrelevant comments degrade it by 34%, confirming gains are tactic-structural

RL Agents for Symbolic Algebra

MIT 6.7920 | *PyTorch, PPO, custom algebra env*

Fall 2025

- Designed custom environment mapping algebraic equation solving to sequential decision-making over AST state spaces with ~ 20 symbolic rewrite rules; tested OOD generalization for PPO, Q-Learning & Sarsa on unseen tasks, showing curriculum benefits for symbolic reasoning depth